

# Practical-3: Gradient Descent Algorithm

**Problem statement:** Implement the gradient descent algorithm to find the local minimum of  $y=(x+3)^2$ , starting from  $x=2$ .

## Steps

1. Define the function and its derivative
2. Initialize parameters
3. Run the gradient descent loop
4. Print the result
5. Plot convergence

## Step 1: Import Libraries

```
import numpy as np
```

## Step 2: Define the function and its derivative

```
def f(x):  
    return (x + 3)**2  
  
def grad_f(x):  
    return 2 * (x + 3)
```

## Step 3: Initialize parameters for gradient descent

```
x_current = 2          # starting point  
learning_rate = 0.1    # step size  
tolerance = 1e-6       # convergence tolerance  
max_iterations = 25    # maximum iterations  
history = [x_current] # storing history
```

## Step 4: Gradient descent loop

```
for i in range(max_iterations):  
    gradient = grad_f(x_current)  
    x_next = x_current - learning_rate * gradient # update step  
  
    # Check convergence  
    if abs(x_next - x_current) < tolerance:  
        print(f"Converged after {i+1} iterations.")  
        break  
  
    x_current = x_next  
    history.append(x_current)
```

```
print(f"Iteration {i+1}: x = {x_current:.4f}, f(x) = {f(x_current):.4f}")
```

```
Iteration 1: x = 1.0000, f(x) = 16.0000
Iteration 2: x = 0.2000, f(x) = 10.2400
Iteration 3: x = -0.4400, f(x) = 6.5536
Iteration 4: x = -0.9520, f(x) = 4.1943
Iteration 5: x = -1.3616, f(x) = 2.6844
Iteration 6: x = -1.6893, f(x) = 1.7180
Iteration 7: x = -1.9514, f(x) = 1.0995
Iteration 8: x = -2.1611, f(x) = 0.7037
Iteration 9: x = -2.3289, f(x) = 0.4504
Iteration 10: x = -2.4631, f(x) = 0.2882
Iteration 11: x = -2.5705, f(x) = 0.1845
Iteration 12: x = -2.6564, f(x) = 0.1181
Iteration 13: x = -2.7251, f(x) = 0.0756
Iteration 14: x = -2.7801, f(x) = 0.0484
Iteration 15: x = -2.8241, f(x) = 0.0309
Iteration 16: x = -2.8593, f(x) = 0.0198
Iteration 17: x = -2.8874, f(x) = 0.0127
Iteration 18: x = -2.9099, f(x) = 0.0081
Iteration 19: x = -2.9279, f(x) = 0.0052
Iteration 20: x = -2.9424, f(x) = 0.0033
Iteration 21: x = -2.9539, f(x) = 0.0021
Iteration 22: x = -2.9631, f(x) = 0.0014
Iteration 23: x = -2.9705, f(x) = 0.0009
Iteration 24: x = -2.9764, f(x) = 0.0006
Iteration 25: x = -2.9811, f(x) = 0.0004
```

## Step 5: Print the result

```
print("Local minima at x =", x_current)
print("Function value at local minima y =", f(x_current))
```

```
Local minima at x = -2.981110534068521
Function value at local minima y = 0.00035681192317650156
```

## Step 6: Plotting

```
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

plt.plot(history, [f(val) for val in history], marker='o')
plt.xlabel("x values")
plt.ylabel("f(x)")
plt.title("Gradient Descent Convergence")
plt.grid()
```

```
plt.savefig("code_3_convergence.png")  
print("Plot saved to code_3_convergence.png")  
Plot saved to code_3_convergence.png
```

